VTPEH 6270 - HW 4

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1 Introduction

1.1 Specific Objective

We will explore:

- Is there a decrease in **reported positive cases of elevated pediatric blood lead level (varA)** over **time in years (varB)** in Nassau, NY? *(27 words)*

Nassau County was selected due to the higher incidences shown in previous iterations of this data exploration.

1.2 Scientific Plausibility and Context

I am interested in the link between geography and blood lead levels in children. Broadly, there is a public campaign to ensure that children are screened for blood lead levels routinely. Due to a variety of social, environmental, and economic factors, children of some backgrounds are more at risk of exposure to lead, which can have devastating long term effects on development and overall health.[1] States typically have standards on providing blood lead testing for children. In the state of New York, all children are supposed to undergo regular checkups to screen for elevated blood lead testing.[2] These guidelines change frequently, as recently as 2019 for medical providers, and with rental registry laws in 2025.[2] Around the country, there have been national efforts to improve drinking water via the 2021 Infrastructure Investment and Jobs Act, these projects are up to the individual states to determine what projects get priority first.[3] For New York State, municipalities with high populations of people of color and lower incomes are likely to be prioritized by the lead service line replacement program, as these individuals may be more at risk for blood lead levels.[3]

References

1. CDC. CDC Updates Blood Lead Reference Value. Childhood Lead Poisoning Prevention. Published April 2, 2024. https://www.cdc.gov/lead-prevention/php/news-features/updates-blood-lead-reference-value.html
2. New York State Department of Health. Childhood Lead Poisoning Prevention. ny.gov. Accessed February 19, 2026. https://www.health.ny.gov/environmental/lead/
3. Lead Service Lines in NYC Disproportionately Impact Hispanic/Latino Communities and Children Already At Risk. Columbia University Mailman School of Public Health. Published August 30, 2023. https://www.publichealth.columbia.edu/news/lead-service-lines-nyc-disproportionately-impact-hispanic-latino-communities-children-already-risk-lead-exposure

2 Materials and Methods

The data set I will be working with is from by the New York State Department of Health. Per state regulation, doctors are required to do blood level testing for all children at specific time intervals. This data includes all children tested for lead from 2000 to March 6, 2024. Important to note, this data does not include information from NYC.

There are nullification qualifications to protect patient anonymity in cases where positive result/number of tests per Zip code is low. Tests are pulled by year, and are reported on the Zip code, geographic center, and county level.

3 Github Accesibility

To see the more information about the data set from the publisher, please visit the website provided: https://health.data.ny.gov/Health/Childhood-Blood-Lead-Testing-and-Elevated-Incidenc/d54z-enu8/about_data

You can access the following dataset and data library at my github links below.

- https://github.com/cal386/VTPEH-repo/blob/main/NYSDOH_Childhood_Blood_Lead_Zip_Code_DataDictionary.pdf
- https://github.com/cal386/VTPEH-repo/blob/main/Childhood_Blood_Lead_Testing_and_Elevated_Incidence_by_Zip_Code__Beginning_2000_20260205.csv

The following code documents the process of cleaning data to get Nassau County level data.

```
# Set up the work directory,

setwd("/Users/clincoln/Desktop/vtpeh-6270")
getwd()

# then load some packages,
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
```

```r
library(gtsummary)
library(gt)
library(viridis)
library(patchwork)

# store the data, and then ...
data_lead <-
  read.csv(
    "Childhood_Blood_Lead_Testing_and_Elevated_Incidence_by_Zip_Code__Beginning_2000_20260205.csv")

# and then preview the data.
head(data_lead)
colnames(data_lead)
summary(data_lead)

# Some of these data are classed as integers, when they should be categories.
data_lead$County.Code <- as.character(data_lead$County.Code)
data_lead$Zip <- as.character(data_lead$Zip)

# And vice-versa
data_lead$Tests <- as.numeric(data_lead$Tests)
data_lead$Less.than.5.mcg.dL <- as.numeric(data_lead$Less.than.5.mcg.dL)

data_lead_tidy <- data_lead

data_lead_tidy <- data_lead %>%
  select(-"Less.than.5.mcg.dL",
         -"X5.10.mcg.dL",
         -"X10...15.mcg.dL",
         -"X15...mcg.dL",
         -"County.Code")

data_lead_tidy <- data_lead_tidy %>%
  rename("Total_Elevated_Blood_Levels" = "Total.Elevated.Blood.Levels.",
         "New_Case_Percentage" = "Percent",
         "Rate_per_1000" = "Rate.per.1.000",
         "ZIP_Location" = "Zip.Code.Location",
         "County_Location" = "County.Location")

# Per my conversation with a peer, I have decided to use
# New Case Percentage in order to reduce bias from the
# quantitative amount of positive tests, and contextaulize
# the score with the amount of tests provided as a proxy for
# Zip-code population size.

# This is actually a defined variable
summary(data_lead_tidy$New_Case_Percentage)

# I'm going to exclude the 0-data...
table(data_lead_tidy$Rate_per_1000)

non0rate_data <- data_lead_tidy[
  data_lead_tidy$Rate_per_1000 > 0,]
```

```r
# and subset for Nassau County
nassau_data <- non0rate_data[
  non0rate_data$County == "Nassau",]

# This should give our new dataset, specifically with NASSAU level data.
summary(nassau_data)
table(nassau_data$Zip)

# and remove old dfs
rm(data_lead, data_lead_tidy, non0rate_data)
```

The following code will generate a table of variables, types, and classes for the `nassau_data`.

```r
# I have to make a new table with the data variables, types, and classes.

data_lead_vars <- data.frame(
  Variable = c("County",
               "Year",
               "Zip",
               "Tests",
               "Total_Elevated_Blood_Levels",
               "New_Case_Percentage",
               "Rate_per_1000",
               "ZIP_Location",
               "County_Location"),
  Type = c("Categorical",
           "Discrete",
           "Categorical",
           "Discrete",
           "Continuous",
           "Continuous",
           "Continuous",
           "Categorical",
           "Categorical"),
  Class = c("Character",
            "Integer",
            "Character",
            "Numeric",
            "Integer",
            "Numeric",
            "Integer",
            "Character",
            "Character")
)

kbl(data_lead_vars[1:9,], booktabs = T,
    caption = "Table with all the variables by their class and type") %>%
  kable_styling(latex_options = c("striped", "scale_down"))
```

```r
# and get rid of it
rm(data_lead_vars)
```

Table 1: Table with all the variables by their class and type

Variable	Type	Class
County	Categorical	Character
Year	Discrete	Integer
Zip	Categorical	Character
Tests	Discrete	Numeric
Total_Elevated_Blood_Levels	Continuous	Integer
New_Case_Percentage	Continuous	Numeric
Rate_per_1000	Continuous	Integer
ZIP_Location	Categorical	Character
County_Location	Categorical	Character

3.1 Statistical Anaylsis

Using the instructions from this assignment, one can use this to simulate some level of data for pediatric blood lead cases in Nassau County, NY starting from 2005. We will start by identifying the slope of the trend line for all grouped ZIP codes, t = 2005-2020.

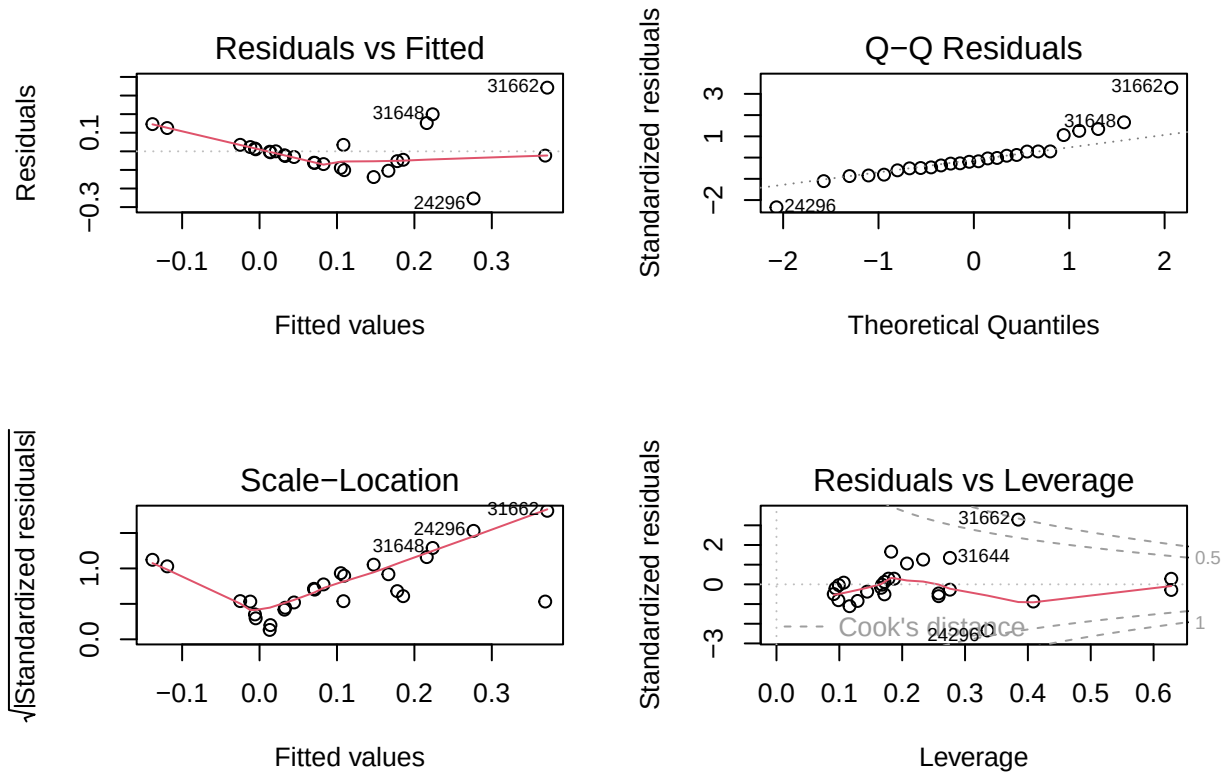
```
# I know that our year data will be categorical
table(nassau_data$Year)

# and our % of new cases/year is continuous
summary(nassau_data$New_Case_Percentage)

# check normality
shapiro.test(nassau_data$New_Case_Percentage)

# run the linear regression
model <- lm(New_Case_Percentage ~ Year + factor(Zip), data = nassau_data)
summary(model)

# check model assumptions
par(mfrow = c(2, 2))
plot(model)
```




```r
# sensitivity analysis excluding 2000, which contained highly influential
# outlying values (range: 34.6-71.4%) that may be distorting model estimates
nassau_no2000 <- nassau_data %>% filter(Year != 2000)
model_no2000 <- lm(New_Case_Percentage ~ Year + factor(Zip), data = nassau_no2000)
summary(model_no2000)
```

Descriptive statistics were calculated for the percentage of newly diagnosed elevated pediatric blood lead levels (New Case Percentage) across Nassau County, NY ZIP codes from 2000 to 2020. The distribution was highly skewed (Median = 1.1%, Mean = 9.38%), driven by elevated values observed in the year 2000 (range: 0.60-71.40%).

A Shapiro-Wilk test indicated significant departure from normality (W = 0.58, $p < 0.001$). Despite this violation, multiple linear regression was selected as the primary analytical approach given that the research question concerns a directional trend over time rather than group differences.

A multiple linear regression was conducted with New Case Percentage as the outcome variable, and Year and ZIP code as predictors (reference category: ZIP 11003). The overall model was statistically significant ($F(6, 20) = 3.93$, $p = 0.009$, Adjusted $R^2 = 0.40$). Year was a significant negative predictor (β = -1.90, SE = 0.48, $p < 0.001$), suggesting a decrease of approximately 1.90 percentage points per year controlling for ZIP code. No ZIP code was a statistically significant predictor (all $p > 0.05$). The full model equation is as follows:

$$\hat{Y} = 3846.57 - 1.90(\text{Year}) - 15.30(\text{ZIP}_{11520}) - 23.37(\text{ZIP}_{11542}) - 14.54(\text{ZIP}_{11550}) + 0.27(\text{ZIP}_{11575}) - 22.25(\text{ZIP}_{11590})$$

Given the presence of influential outlying values in the year 2000, a sensitivity analysis was conducted excluding 2000 observations. The overall model was no longer statistically significant ($F(6, 16) = 1.53$, p =

0.23, Adjusted R^2 = 0.13); however, Year remained a significant negative predictor (β = -0.56, SE = 0.20, p = 0.013), suggesting a modest decrease of approximately 0.56 percentage points per year independent of the 2000 outliers. The sensitivity model equation is as follows:

$$\hat{Y} = 1140.21 - 0.56(\text{Year}) - 3.89(\text{ZIP}_{11520}) - 6.89(\text{ZIP}_{11542}) - 4.48(\text{ZIP}_{11550}) - 5.07(\text{ZIP}_{11575}) - 5.78(\text{ZIP}_{11590})$$

Several limitations should be noted. First, the data violated the normality assumption of linear regression. Second, many year-ZIP combinations had only one observation, limiting statistical power. Third, the year 2000 contained highly influential outlying values that substantially affected model estimates, as demonstrated by the sensitivity analysis. Finally, missing data across years and ZIP codes may introduce bias into the estimates.

4 Results

4.1 Data Visualization

```r
na.omit(nassau_data) %>%
  group_by(Zip) %>%
  summarise(
    median = median(New_Case_Percentage, na.rm = TRUE),
    q1 = quantile(New_Case_Percentage, 0.25, na.rm = TRUE),
    q3 = quantile(New_Case_Percentage, 0.75, na.rm = TRUE),
    mean = mean(New_Case_Percentage, na.rm = TRUE),
    sd = sd(New_Case_Percentage, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    `Median (Q1, Q3)` = sprintf(
      "%.1f (%.1f, %.1f)",
      median, q1, q3
    ),
    `Mean (SD)` = sprintf(
      "%.1f (%.1f)",
      mean, sd
    )
  ) %>%
  select(Zip, `Median (Q1, Q3)`, `Mean (SD)`) %>%
  gt() %>%
  tab_header(
    title = md("**Annual Percentage of New Cases of Pediatric Blood Lead Tests
               Reported in Nassau County, NY
               2000-2020**"),
    subtitle = "ZIP Code summary statistics"
  ) %>%
  tab_options(
    table.font.size = px(11),
    table.width = pct(60),
    heading.title.font.size = px(12),
    heading.subtitle.font.size = px(11),
    column_labels.font.size = px(11)
  )
```

```r
nassau_tidy <- nassau_data %>%
  filter(!is.na(Zip),
         !is.na(Year),
         !is.na(New_Case_Percentage))
rm(nassau_data)

nassau_tidy$New_Case_Percentage <- nassau_tidy$New_Case_Percentage*100

p1 <- ggplot(nassau_tidy, aes(x = factor(Year),
                      y = Zip,
                      fill = New_Case_Percentage)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(New_Case_Percentage, 1),
             color = factor(New_Case_Percentage > 20)),
         size = 3,
         show.legend = FALSE) +
  scale_color_manual(values = c("TRUE" = "black", "FALSE" = "white")) +
  scale_fill_viridis(option = "C", name = "% Cases") +
  labs(
    title = "Percent of Newly Diagnosed Pediatric Elevated Blood Lead Levels",
    subtitle = "Nassau County, NY (2000-2020)",
    x = "Year",
    y = "ZIP Code"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 7),
    axis.text.y = element_text(size = 9),
    plot.title = element_text(face = "bold")
  )

p2 <- ggplot(nassau_tidy, aes(x = Year,
                        y = New_Case_Percentage,
                        group = Zip,
                        color = Zip)) +
  geom_line(alpha = 0.95, linewidth = 0.7) +
  geom_point(size = 2) +
  geom_smooth(
    data = subset(nassau_tidy, Year >= 2005),
    aes(group = 1, linetype = "Trend (2005-2020)"),
    method = "lm",
    color = "black",
    size = 0.7,
    se = TRUE
  ) +
  scale_color_viridis_d(option = "C", name = "ZIP Code") +
  scale_linetype_manual(name = "", values = "dashed") +
  labs(
    title = "Percent of Newly Diagnosed Pediatric Elevated Blood Lead Levels",
    subtitle = "Nassau County, NY (2000-2020)",
    x = "Year",
    y = "Percentage of New Cases (%)"
  ) +
```

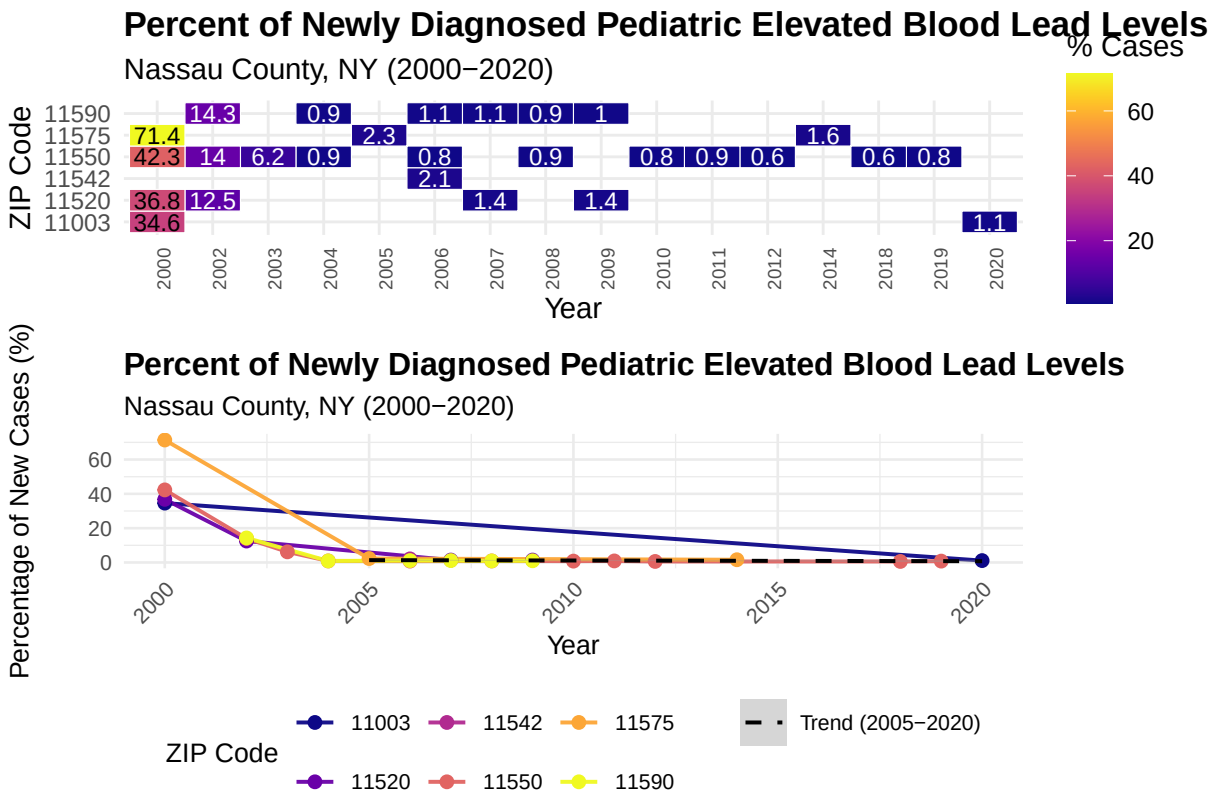
Annual Percentage of New Cases of Pediatric Blood Lead Tests Reported in Nassau County, NY 2000-2020

ZIP Code summary statistics

Zip	Median (Q1, Q3)	Mean (SD)
11003	0.2 (0.1, 0.3)	0.2 (0.2)
11520	0.1 (0.0, 0.2)	0.1 (0.2)
11542	0.0 (0.0, 0.0)	0.0 (NA)
11550	0.0 (0.0, 0.1)	0.1 (0.2)
11575	0.0 (0.0, 0.4)	0.3 (0.4)
11590	0.0 (0.0, 0.0)	0.0 (0.1)

```
  theme_minimal(base_size = 10) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom",
    plot.title = element_text(face = "bold")
  )

p1/p2
```



4.2 Statistical Analysis

```r
library(gtsummary)

# full model table
tbl_full <- tbl_regression(model,
  label = list(
    Year ~ "Year",
    `factor(Zip)` ~ "ZIP Code"
  )) %>%
  add_glance_source_note(
    include = c(r.squared, adj.r.squared, statistic, p.value)
  ) %>%
  modify_header(label = "**Predictor**") %>%
  modify_caption("**Full Model (2000-2020)**")

# sensitivity model table
tbl_sens <- tbl_regression(model_no2000,
  label = list(
    Year ~ "Year",
    `factor(Zip)` ~ "ZIP Code"
  )) %>%
  add_glance_source_note(
    include = c(r.squared, adj.r.squared, statistic, p.value)
  ) %>%
  modify_header(label = "**Predictor**") %>%
  modify_caption("**Sensitivity Model (2002-2020)**")

# merge both tables side by side
tbl_merge(
  tbls = list(tbl_full, tbl_sens),
  tab_spanner = c("**Full Model (2000-2020)**",
                  "**Sensitivity Model (2002-2020)**")
)
```

4.3 Conclusion

This analysis examined whether there was a statistically significant decrease in the percentage of newly diagnosed elevated pediatric blood lead levels in Nassau County, NY between 2000 and 2020. Using multiple linear regression controlling for ZIP code, results support a statistically significant downward trend over time.

In the full model, Year was a significant negative predictor of new case percentage (β = -1.90, SE = 0.48, p $<$ 0.001), suggesting a decrease of approximately 1.90 percentage points per year. However, this estimate was heavily influenced by outlying values in the year 2000, where two ZIP codes (11575 and 11550) reported case percentages of 71.4% and 42.3% respectively — substantially higher than all subsequent years.

A sensitivity analysis excluding 2000 revealed a more modest but still statistically significant downward trend (β = -0.56, SE = 0.20, p = 0.013), supporting the conclusion that elevated pediatric blood lead levels in Nassau County declined over the study period, independent of the influence of early outlying values. ZIP code was not a statistically significant predictor in either model, suggesting that the temporal trend was consistent across geographic areas within the county.

Table 2: **Full Model (2000-2020)**

	Full Model (2000-2020)			Sensitivity Model (2002-2020)		
Predictor	**Beta**	**95% CI**	**p-value**	**Beta**	**95% CI**	**p-value**
Year	-0.02	-0.03, -0.01	<0.001	-0.01	-0.01, 0.00	0.013
ZIP Code						
11003	—	—		—	—	
11520	-0.15	-0.40, 0.09	0.2	-0.04	-0.16, 0.08	0.5
11542	-0.23	-0.58, 0.11	0.2	-0.07	-0.21, 0.07	0.3
11550	-0.15	-0.36, 0.07	0.2	-0.04	-0.15, 0.06	0.4
11575	0.00	-0.25, 0.26	>0.9	-0.05	-0.17, 0.06	0.4
11590	-0.22	-0.45, 0.01	0.057	-0.06	-0.17, 0.05	0.3

Abbreviation: CI = Confidence Interval
$R^2 = 0.364$; Adjusted $R^2 = 0.126$; Statistic = 1.53; p-value = 0.2
$R^2 = 0.541$; Adjusted $R^2 = 0.403$; Statistic = 3.93; p-value = 0.009

These findings are consistent with broader public health efforts to reduce childhood lead exposure, including mandatory screening programs in New York State and national infrastructure investments targeting lead service line replacement. However, several limitations temper these conclusions, including violation of the normality assumption, sparse data in later years, and the disproportionate influence of year 2000 observations on model estimates. Future analyses with more complete data across years and ZIP codes would strengthen these findings.

5 AI disclosure statement

The author used Claude (Anthropic, claude.ai) as an AI assistant during the preparation of this analysis. AI assistance was used for the following purposes: debugging R code and checking interpretations for statistical output. All statistical analyses were performed in R, and written by me.